CYBERJAYA 5G FUTURE TALENT INNOVATION CHALLENGE 2026

Track 2 · 5G Connected & Safe Last-Mile Mobility

MATA KEDUA

A second pair of eyes for Cyberjaya's driverless vans

When a driverless van hesitates near a school gate or a flooded road, a remote operator gets a second view over 5G and decides what it does next.

Team **Bob** · **Asia Pacific University**

Organised by

KOLAXUS
Business Innovation Platform

In collaboration with

CYBERVIEW

Team Information

Team Name
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Team Members (3)

Name	Position
Chan Hao Wen	Team Leader
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1. The Problem

When Cyberjaya's driverless van asks for help, its operator sees only the van's own cameras.

7:15 am, a school gate near the van's route. Cars are double-parked. The van stops behind a tall SUV and asks the control room: go round it? The operator sees the back of the SUV, so "Go" looks safe. A child is about to step out.
Illustrative scene

23 Jan 2026 Santa Monica: a driverless Waymo struck a 9-year-old who ran out from behind a double-parked SUV at school drop-off (minor injuries).

67 a year deaths in Malaysian school zones, all road users (police data).

RTA s.79 an officer's hand signal overrides the traffic light, so a driverless van must obey a person.

- **On the road since 20 Oct 2026:** ALS's fully driverless RoboVan, on a six-month trial past educational institutions, MRT stations and crossings.
- **Edge cases on its route:** the Persiaran Rimba Permai junction residents call "simpang maut", where the council posted officers to direct traffic. The same road flooded in 2019.

ON RECORD IN MALAYSIA

26 Oct 2026 An 11-year-old was seriously hurt on the crossing in front of a school in Kota Kinabalu, at 6:10 am.

22 Oct 2026 At the first Safe School Zone pilot school, 76.6% of pupils said they felt unsafe on the roads around it.

May 2024 Parents double-park in the middle of the road at school gates; some schools now run one-way systems.

Sources: NHTSA PE26001; NTSB HWY26FH008; WHO 4 Aug 2024 (police data); Road Transport Act 1987 s.79; NST 20 Oct 2026; Utusan 29 Nov 2025; Malaysia Tribune 1 Dec 2025; The Sun 2019; Berita Harian 22 and 26 Oct 2026; NST 2 May 2024. Links in appendix.

2. Target User

Four groups depend on a decision the remote operator makes in seconds.

People around the van (who it protects)

Children and parents at drop-off, school guards, riders filtering past, people at MRT crossings. They can't tell what a stopped van will do, so they go round it in the live lane.

The remote operator (who uses MATA KEDUA)

At the operations centre run by ALS with Sidec and Cyberview. Sees the van's own video, speed, position and status; must decide in seconds from angles that can't see past a parked SUV.

The van's engineers (ALS, Zelostech)

Need Malaysian edge cases to retrain the van; today they surface one intervention at a time.

Futurise, MOT, MIROS

Need evidence for Malaysia's first AV rules; the trial guideline doesn't cover remote operation.

INSIGHT

Driverless vans rarely fail dramatically. They hesitate, in Malaysian situations nobody trained them for: a guard's hand signal, a child behind a double-parked car, water across the road.

A person can resolve most of these in seconds with a better view, and that decision should be recorded.

Abroad: a remote agent misread a school bus (Jan 2026).

Ipsos 2026: 6 in 10 Malaysians would feel safe in a self-driving car.

Sources: Business Today 31 Jul 2026 (Remote Operations Centre); Futurise AV Trial Guideline v6.0; NTSB HWY26FH007 (Jan 2026); Ipsos Mobility Monitor 2026.

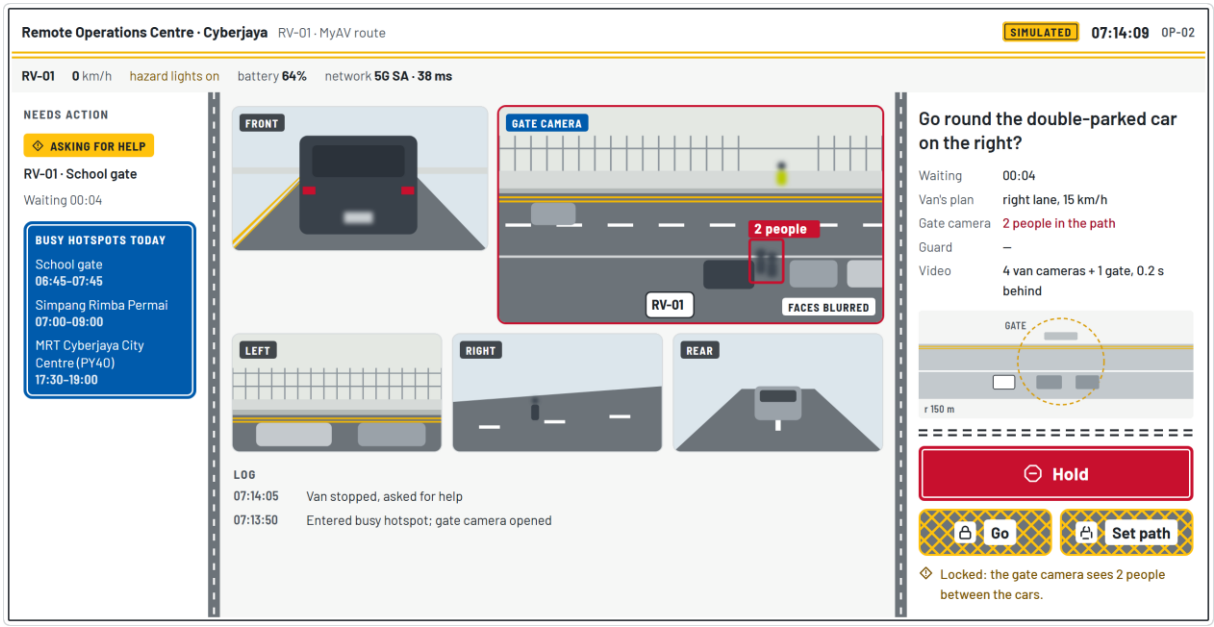
3. Our Solution

Three additions to the control room ALS already runs.

1 More to see when the van asks. The van stops safely and sends four camera views and what its sensors detect over a prioritised 5G link. The operator answers Hold, Go or Set path.

2 A roadside view at hotspots. At a school gate, a junction with officers directing traffic or a flood dip, the roadside camera's view opens when a van there asks or approaches in busy hours. Go stays locked while it shows people in the way.

3 A label for every assist. Place, time, type, decision and a blurred clip go into a Malaysian edge-case library that ALS can retrain the van with.



07:14:05 · the gate camera sees 2 people; Go and Set path are locked

The operator never steers. The van checks every path with its own sensors and can refuse it.

Screenshot from our interactive prototype (simulated run; camera views are drawings).

4. Innovation

Remote help and roadside views already exist abroad; three things are new in Malaysia.

Already exists	New in MATA KEDUA
ALS's operations centre sees the van's own video, speed, position and status. ALS logs interventions; Futurise requires monthly near-miss reports.	**Roadside views at Malaysian hotspots,** opened when a van asks or approaches in busy hours. *No Malaysian precedent found (our search, Oct 2026).*
Remote help that advises and never steers: Waymo (about 70 agents on duty worldwide for about 3,000 cars), Pony.ai (1 per 20 robotaxis)	**A shared Malaysian edge-case label set,** with a blurred roadside clip per assist, for operators and regulators
Roadside cameras feeding remote monitors in Japan (Eiheiji 2023; Komae over local 5G, 2024–25)	**The first measured 4G vs 5G upload map** of the MyAV route, stop by stop
5G remote monitoring of an autonomous bus in Cyberjaya (DNB, Ericsson, eMooVit, 2024)	*Design lessons: Hold is the default when views disagree; staffing follows the hotspot map*

Different from the briefing's "a street detects a cyclist before a driver does": our roadside camera warns no driver or child. It gives one remote person a second view, so they can decide for the van. Japan shows the pattern works; we bring it to Malaysian hazards.

Sources: ALS and NST; Futurise AV guideline; Waymo; Pony.ai; StVG §1f; RoAD to the L4 (Eiheiji); MRI/MIC (Komae); DNB; Cyberview. Links in appendix.

5. How it Works

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One assist at a school gate takes 38 seconds.

Time	What happens	Link
7:13:50	In the gate's busy hours the van approaches at 30 km/h; the gate camera's view opens for the operator (faces blurred on the pole).	5G from the pole
7:14:05	A car is double-parked in its lane. The van stops behind it and asks: go round on the right? Go is locked: the gate camera sees 2 people.	5G, priority
7:14:10	Without the gate view, "Go" would look safe. The operator presses Hold and asks the guard to confirm.	—
7:14:30	The guard walks the pupils across and holds "clear" on the gate phone. The gate camera shows 0 people in the path.	Short message
7:14:34	The operator sets a path: right lane, 10 km/h. The van checks it with its own sensors and moves.	Short message
7:14:43	Done in 38 s. The operator confirms the labels: school gate · double-parking · pupils crossing · guard signalling.	—

False alarm: both views clear, the operator presses Go. **Frozen video:** it stays Hold and ALS's ground team is called.

Same timeline as the prototype (simulated). The first live site would be Simpang Rimba Permai (on the route); school gates only in Phase 3, with the school's agreement.

5. How it Works: Why 5G?

KOLAXUS Business Innovation Platform CYBERVIEW

Four views need 16–32 Mbps of upload; Malaysia's 5G averages about 32, all networks about 16.

Remote help can wait a moment (Waymo's assistants answer from 150–250 ms away); the limit is upload room at the busiest moment.

What the operator needs	Evidence	Without 5G
Four views from one van	4–8 Mbps per HD view (Ericsson). Average upload, four largest operators, Apr–Jun 2026: ~32 Mbps on 5G, 15 to 17 Mbps across their networks; 2 to 7 Mbps in the 4G-only era (Opensignal).	**One or two lower-quality views at peak: not enough to see round a parked SUV.**
A reserved lane at 7:15 am	Priority is sold on 4G and 5G but is best-effort. Only 5G Standalone can reserve a guaranteed lane (a slice), so we need a trial with a telco. Even uncrowded 5G SA dropped 3.6–13% of frames from three merged cameras (arXiv 2507.20438).	**The help request competes with every parent's phone at the gate.**
Room for the roadside view	The gate camera's view goes through the same crowded cell at the same moment. Japan's Komae project uses local 5G for this.	**The roadside view and the van's views squeeze into a full 4G cell.**

What 4G can already do: on a quiet cell, one or two compressed views, enough for a yes/no decision on an empty road.

Near the route (Ookla, Jan–Jun 2026): median upload 39 Mbps in Putrajaya, 22 across Selangor. Our estimate at 8 Mbps per HD view: four views fit in Putrajaya's median, two in Selangor's. Phase 1 measures the MyAV route itself.

Sources: Ericsson 2017 (LTE test); Opensignal Jul 2026 and Sep 2022 (averages); Ookla H1 2026 (medians); arXiv 2507.20438; Maxis; U Mobile; MRI/MIC Komae; Waymo 17 Feb 2026.

6. Impact

We will measure people held for, assist time and lane-blocking minutes at 5 hotspots.

Who benefits

- **Children, parents, guards, riders:** a van that waits when someone is hidden and moves on when the road is clear
- **Operators:** decisions from a better view
- **ALS:** quicker assists, Malaysian training data, a network plan
- **Futurise, MOT, MIROS:** evidence for Malaysia's first AV rules
- **Cyberview:** a Smart Mobility showcase on its route

Fits national priorities: the Road Safety Council's Safe School Zone pilot (14 schools, launched 22 Oct 2026) and the 30 km/h school-zone limit now being gazetted.

Pilot metric	Phase	Target
Upload per stop, 4G vs 5G, typical and the slowest 1 in 10	1	≥ 20 stops × 3 times of day
Stops where four views plus a roadside view stream without stalling	1	Measured, 4G vs 5G
Hotspots mapped with peak-time counts	1	5
Times the operator held the van for someone it couldn't see	2–3	Every one logged
Time to resolve an assist (typical and slowest 1 in 10)	2–3	Baseline, then a measured drop
Minutes the van blocks a lane at hotspots	2–3	A measured drop
Wrong confirmations caught by the van's own check	2–3	Every one logged

Sources: Pilot metrics are ours. Road Safety Council pilot: NST 15 Jun 2026 and The Star 22 Oct 2026.

7. Feasibility

Phase 1 costs about RM 5,000 and needs no one's permission.

Partners

- **ALS (with Zelostech), Sidec and Cyberview:** the operations centre they launched in Jul 2026
- **Futurise:** the MyAV route and the regulatory sandbox
- **MPSepang:** junction cameras, e.g. Simpang Rimba Permai
- **A telco:** a priority or slicing trial (U Mobile or Yes run 5G Standalone)
- **MIROS:** independent evaluation

Letter of interest

Next step: we ask ALS and Cyberview for a letter of interest in a Phase 2 shadow trial.

Phase 1 budget (public prices; to confirm with quotes)

Item	RM
Industrial 5G router, two SIM slots	2,690
Data, 3 months: two lines, one on 5G Standalone	450
Priority add-on during tests (RM 2 per hour, ~30 hours)	60
Small cloud server to receive the video	150
Car mount, portable power, cables	600
Fuel and parking, about 15 drives	400
Hotspot counts and survey thank-yous	300
Contingency (about 10%)	450
Total	**5,100**

Our laptop and phones stream recorded road footage, so no one is filmed. Funding ask: Cyberview or Futurise support, or an APU grant.

Sources: Public list prices, Oct 2026 (Cytron Malaysia; thewalaoeh plan table; U Mobile). To confirm with quotes.

7. Feasibility: Safety and Privacy

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Live view only when a van needs it, one blurred clip per assist, and no remote driving.

Safety

- **Hold, Go or Set path only.** The van checks every path and can refuse it, as German law requires of remote supervisors.
- **A safe stop is the default.** Frozen video, disagreeing views or no answer: the van stays stopped and ALS's ground team is called.
- **Go locks while people are in view.** A guard's "clear" is one more signal, never enough alone.
- **Staffing follows the hotspot map,** with extra cover at 7:00–7:45 am and in monsoon storms.

Privacy: PDPA, and cameras near children

- **Blurred on the pole.** Faces and number plates are blurred before video leaves the camera. No face recognition, no audio.
- **Live only when needed:** a van within 150 m is asking, or approaching in registered busy hours. Otherwise nothing is sent.
- **One short record.** A blurred 30-second clip per logged assist, deleted after 30 days unless police or Futurise ask.
- **Signs at every camera; school gates last:** Phase 3 only, with the school's and PIBG's agreement.

We will not drive the van remotely, recognise faces, or send warnings to children or drivers.

Sources: PDPA 2010 as amended by Act A1727 (2024): biometric data is sensitive personal data; JPDP consultation on CCTV (2025); German StVG §1f; NTSB HWY26FH007.

8. Implementation

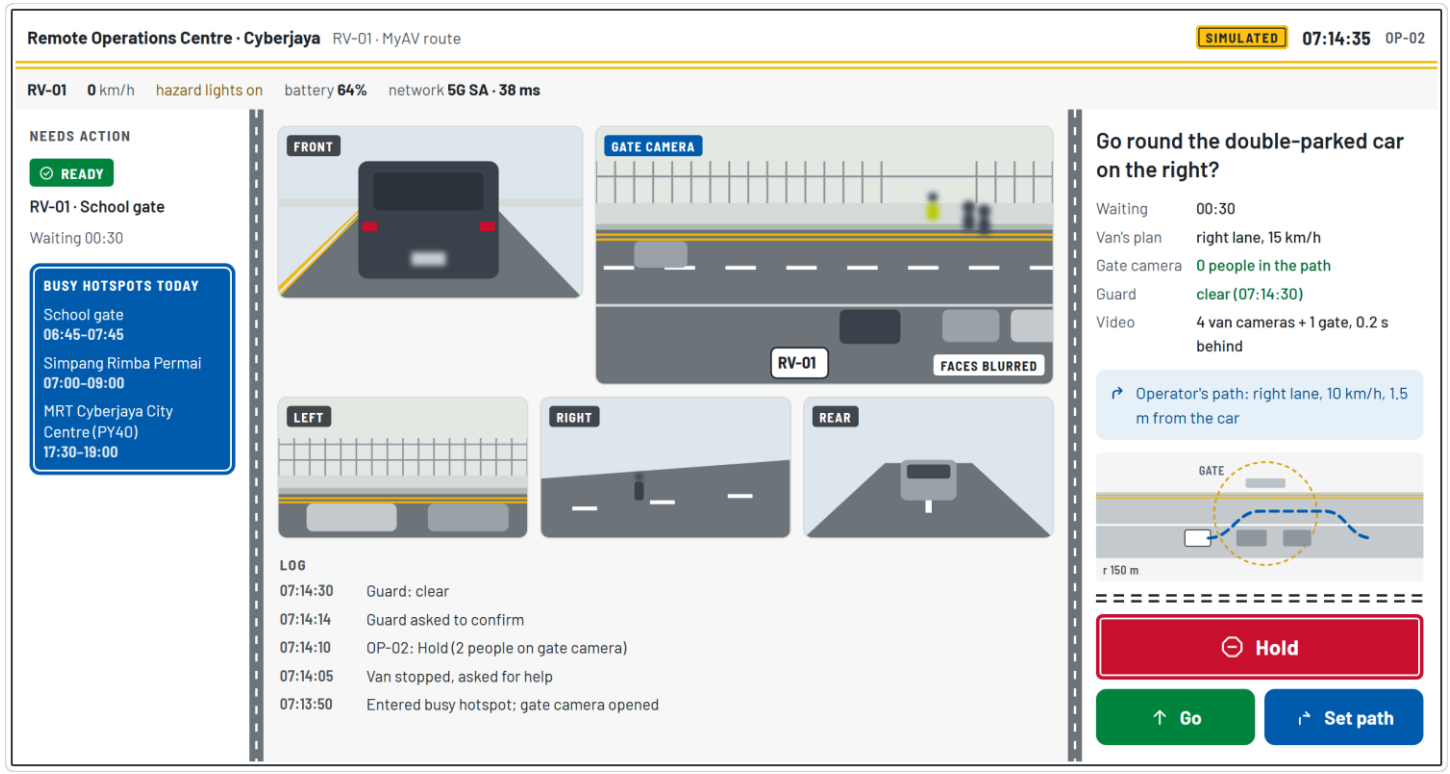
Four phases, each with a go/no-go check before the next.

Phase	What	Go / no-go check
0 Desk work Oct 2026	Sources checked, remote help abroad studied, interactive prototype built	Done
1 Map it Nov 2026 – Jan 2027 no permission needed	A laptop and 5G router in a car stream four recorded views at 20+ stops, 3 times of day; a phone at each hotspot streams a fifth recorded view. Counts at 5 hotspots (monsoon included) and a resident survey.	Go if 5G carries five views at most hotspots at peak. If 4G proves enough, ALS saves money and we publish that.
2 Shadow mode Feb – Jul 2027 an offer to ALS, Sidec, Cyberview, Futurise	At 1–2 non-school hotspots (e.g. Simpang Rimba Permai), the roadside view shows on a separate screen when the van is there. The van doesn't change. Priority trial with a telco. One roadside unit: about RM 7,800 in parts.	Go if ALS and the sandbox agree, operators rate the view useful in half of hotspot assists, and a privacy review passes.
3 Live use Aug 2027 – Jan 2028	The view joins ALS's console. First school gate, with the guard's phone, only with the school, PIBG, MPSepang and the Road Safety Council.	Go if wrong "Go"s caught, assist time and lane-blocking minutes improve, and community trust holds.
4 Scale 2028+	Other MyAV routes and AV operators; the labels feed Malaysia's AV roadmap	More operators adopt it

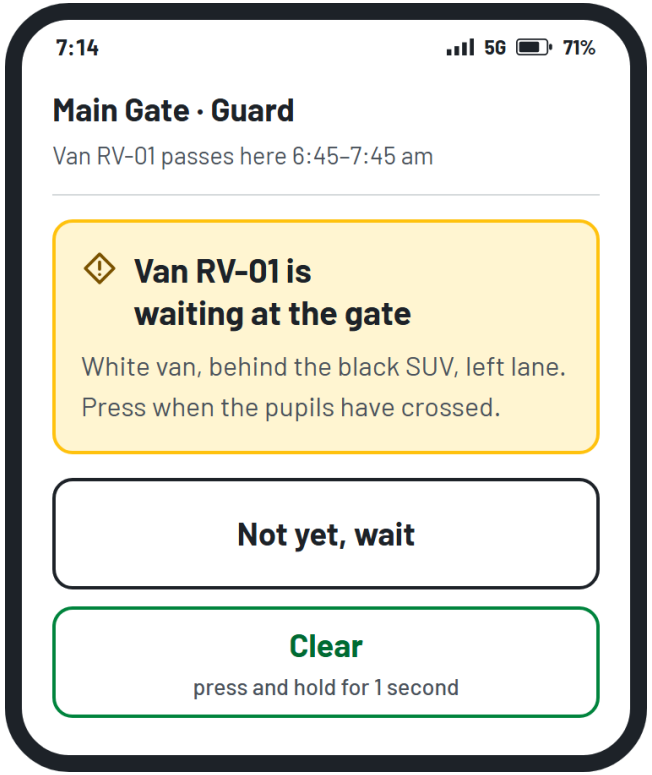
Anything a regulator or a school must approve (public-road cameras, joining ALS's systems, school-gate views) starts in Phase 2 or later.

9. Visual Concept

Our interactive prototype: the operator's console and the guard's phone.



07:14:34 · gate camera clear; the operator sets a path

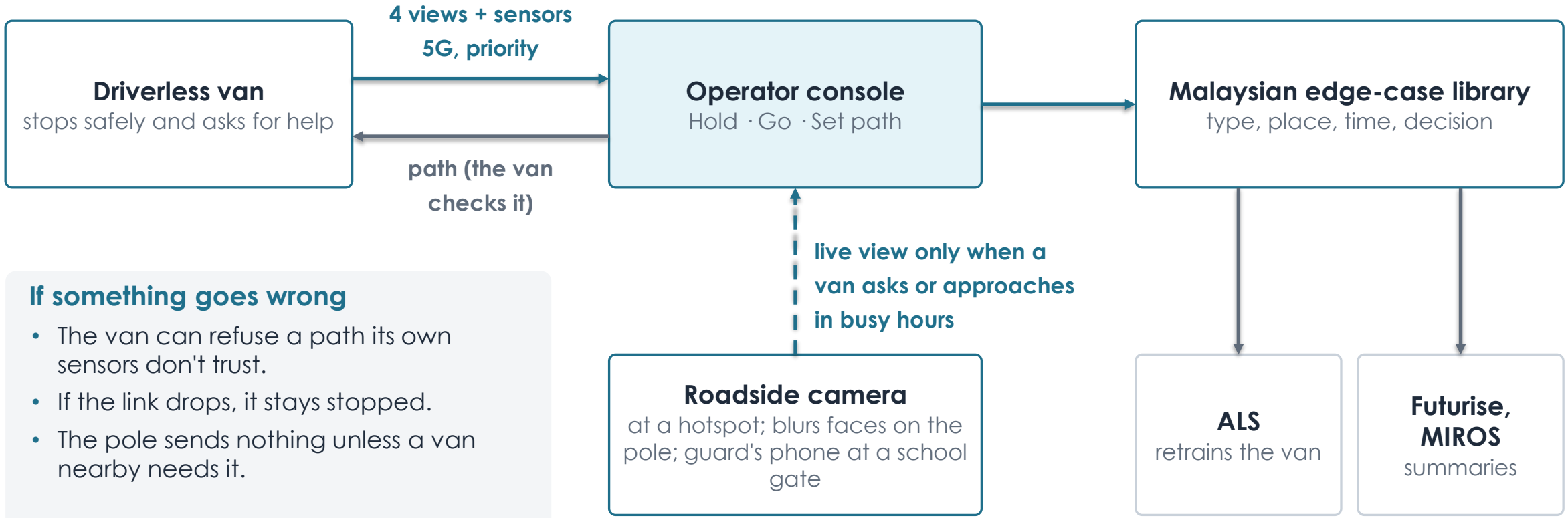


Guard's phone (Phase 3)

Try it: https://zghanw.github.io/matakedua/ (A simulated run plays on its own; click anything to take over)

Screenshots from our interactive prototype (simulated run; camera views are drawings). It also has an edge-case library view for ALS's engineers.

9. Visual Concept: How It Connects



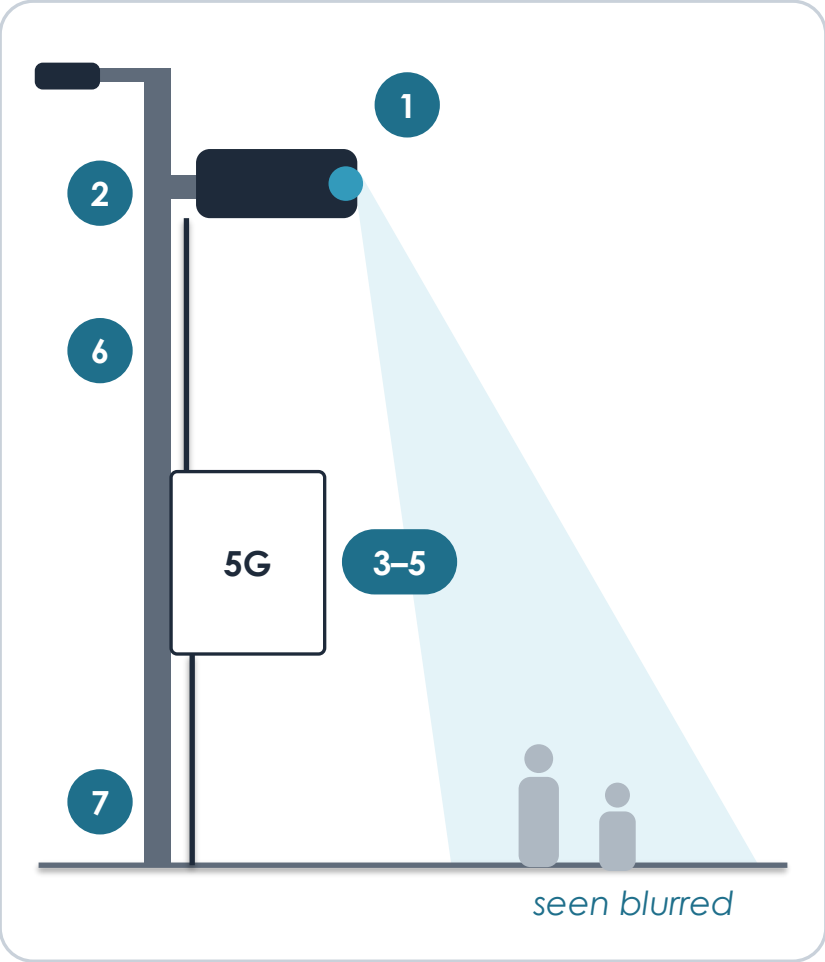
If something goes wrong

- The van can refuse a path its own sensors don't trust.
- If the link drops, it stays stopped.
- The pole sends nothing unless a van nearby needs it.

Solid arrows: every assist. Dashed: only at a registered hotspot, only while a van there is asking or approaching in busy hours.

9. Visual Concept: The Roadside Unit

From Phase 2, each hotspot gets one roadside unit: about RM 7,800 in parts.



#	Part	RM
1	Camera that blurs faces and plates itself (AXIS P1465-LE)	3,430
2	Stainless steel pole mount (AXIS T91B47)	430
3	5G router, two SIMs, SIRIM-approved (Teltonika RUTX50)	2,690
4	24 V supply for both, rated to 70 °C (Mean Well HDR-60-24)	220
5	IP66 plastic box, so the 5G signal gets through	30
6	Cables, glands, surge protector (our allowance)	300
7	*Installation and mains power from the pole*	*quote*
	Contingency (about 10%)	700
	Total per hotspot, parts only	**7,800**

- **Not in the Phase 1 budget (RM 5,100):** there, a phone stands in for the camera. Phase 2 needs 1–2 units.
- **Running cost:** about RM 200 a month for a business 5G line with priority. Item 7: the pole owner's quote.
- **Limits:** blurring runs at 5–10 frames a second and can miss distant people, so each camera covers its crossing up close. Its unblurred stream stays off.

Sources: list prices, Oct 2026 (Unisol, Ingram in US$ at BNM's 4.0818; Cytron, RS, Sierra-Plus); Axis and Teltonika documentation. To confirm with quotes.

10. Why Our Team Should Win

KOLAXUS Business Innovation Platform CYBERVIEW

A driverless van now shares Cyberjaya's roads with children, riders and officers waving traffic through. We did the legwork: we checked our key figures against their sources, studied how remote help works abroad and built an interactive prototype in Bahasa Melayu and English. New for Malaysia: roadside views at local hotspots, a shared edge-case label set and a measured upload map. Phase 1 costs about RM 5,000 and can start now. The van stops safely; a person who can see decides.

Bila van ragu, manusia nampak lebih jauh.

Appendix: AI Usage and Sources

AI usage: we used AI tools to search and summarise sources, tighten our wording, review our draft as a mock judge and help build the interactive prototype. We checked key figures against the original pages. The concept and all design decisions are our own.

- NHTSA PE26001, Waymo struck a child near a school (2026)
- NTSB HWY26FH008, Santa Monica crash (23 Jan 2026)
- NBC Los Angeles, Waymo statement (29 Jan 2026)
- NST, RoboVan trial (20 Oct 2026)
- Business Today, operations centre (31 Jul 2026)
- WHO, school-zone deaths (4 Aug 2024)
- The Star, 30 km/h being gazetted (22 Oct 2026)
- NST, Safe School Zone pilot (15 Jun 2026)
- Road Transport Act 1987, s.79
- Utusan, simpang maut Rimba Permai (29 Nov 2025)
- Malaysia Tribune, MPSepang measures (1 Dec 2025)
- The Sun, Cyberjaya flash flood (May 2019)
- Futurise, AV Trial Guideline v6.0
- Waymo, "Advice, not control" (17 Feb 2026)
- Waymo, San Francisco outage (23 Dec 2025)
- NTSB HWY26FH007, school bus (2026)
- Pony.ai Q1 2025 earnings call (1:20)
- German StVG §1f, technical supervisor
- RoAD to the L4, Eiheiji remote room (2023)
- MRI/MIC, Komae local-5G project (2024)
- DNB, 5G autonomous bus demo (9 Sep 2024)
- Cyberview, smart city (LTE traffic cameras)

Appendix: Sources (continued)

- Ericsson Mobility Report (2017 LTE test)
- arXiv 2507.20438, cameras over 5G SA (2025)
- Opensignal Malaysia (Jul 2026)
- Opensignal Malaysia, 4G era (Sep 2022)
- Opensignal, U Mobile 5G SA (6 Aug 2026)
- YTL / Yes, 5G-Advanced and slicing
- Maxis Business Priority Pass (4G and 5G)
- U Mobile ULTRA5G Priority Pass
- PDPA amendment, Act A1727 (2024)
- JPDP consultation on CCTV (2025)
- Prices: Teltonika RUTX50 (Cytron Malaysia)
- Prices: Malaysian postpaid plans (Sep 2026)
- Berita Harian, pupil hurt at a school crossing (26 Oct 2026)
- Berita Harian, Safe School Zone pilot launch (22 Oct 2026)
- NST, double-parking at school gates (2 May 2024)
- Ookla, Malaysia Speedtest report H1 2026
- Ipsos Mobility Monitor 2026
- Axis, Live Privacy Shield (faces and plates)
- Axis, privacy white paper (up to 10 fps)
- Axis, Live Privacy Shield manual (range)
- Prices: AXIS P1465-LE camera (Unisol)
- Prices: AXIS T91B47 pole mount (Ingram)
- Prices: Mean Well HDR-60-24 (RS Malaysia)
- Prices: IP66 boxes (Sierra-Plus)
- Teltonika RUTX50 power options
- Bank Negara, USD/MYR reference rate